create cloud computing

Cloud Computing 1. Introduction - Cloud Computing is a method of delivering computing services over the internet. - It allows users to access and use computing resources, such as servers, storage, and applications, on-demand. 2. Types of Cloud Computing - Public Cloud: services provided by third-party providers over the internet - Private Cloud: services provided within an organization's internal network - Hybrid Cloud: combination of public and private cloud services 3. Characteristics - Scalability: ability to scale up or down to meet changing needs - On-demand Self-service: users can provision and manage resources on-demand - Broad Network Access: resources accessible over the internet - Resource Pooling: resources are pooled together to provide a multi-tenant environment - Rapid Elasticity: ability to quickly provision and de-provision resources 4. Benefits - Cost Savings: reduced capital and operational expenses - Increased Flexibility: ability to quickly deploy and scale applications - Improved Reliability: built-in redundancy and failover capabilities - Enhanced Security: robust security features and compliance with industry standards 5. Service Models - IaaS (Infrastructure as a Service): provides virtualized computing resources - PaaS (Platform as a Service): provides a platform for developing and deploying applications - SaaS (Software as a Service): provides software applications over the internet 6. Deployment Models - Public Cloud: suitable for public-facing applications and services - Private Cloud: suitable for sensitive and mission-critical applications - Hybrid Cloud: suitable for applications that require both public and private cloud services 7. Key Terms - Virtualization: creation of virtual computing resources - Containerization: packaging of applications and their dependencies - Cloud Storage: storage of data in a cloud environment - Cloud Security: protection of cloud resources and data from unauthorized access.